

Roll No.: 22030102023

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Compiler Design

Course Code: CSPC-613

Examination: 2nd Periodical Examination

Scheme: NEP

Branch: CSE (AI & ML)

Semester: VI

Question Paper for 2nd Periodical Examination (May 2026) (Session Jan-June 2026)

Time: One and a half hours

Max Marks:30

Note: Attempt all questions.

Date: 05.05.2026(10-11:30AM)

CO	COURSE OUTCOMES
CO1	To teach the concept of language translation and the phases of compiler design.
CO2	To inculcate the knowledge of the common forms of parsers, the LL parser and the LR parser.
CO3	To demonstrate intermediate code using the technique of syntax-directed translation.
CO4	To illustrate the various optimization techniques for designing optimizing compilers.

Q No.	Questions	Marks	Target CO	Target KL
1	Answer the following questions:	2*5=10	1,2,3	
1.a	Differentiate LR(0) and LR(1) items with an example.	2	CO3	L3
1.b	Differentiate CLR and LALR parsers.	2	CO3	L3
1.c	What are syntax-directed definitions?	2	CO4	L2
1.d	Differentiate loop fission and loop splitting.	2	CO4	L3
1.e	Write rules to form a basic block and a flow graph with an example.	2	CO4	L2
2.	Explain YACC and its working.	5	CO2	L2
3.	Write DAG representation of a basic block.	5	CO4	L3
4	Implement LALR parser with example.	5	CO3	L3
5.	Explain three address codes with all forms.	5	CO4	L3

KL: Knowledge Level: L1 - Remember L2 - Understand L3 - Apply L4 - Analyze L5 - Evaluate L6 - Create

Course Advisor: Shivani (Assistant Professor, CSE)

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26

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